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A physiological crisis drives the coral recruitment bottleneck

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Recruitment failure is an important factor contributing to population declines of tropical corals. Because the causes of death for juvenile corals are unclear, it is challenging to predict how recruitment bottlenecks will change in the future. We tested the hypothesis that depletion of metabolic reserves increases mortality of juvenile corals under thermal stress. Metabolic reserves of juvenile colonies (<30 mm diameter) of broadcast spawning *Pocillopora* from Moorea, French Polynesia, were manipulated using elevated temperature to increase respiration, and reduced day length to decrease photosynthesis, and estimated as biomass. Corals with high or low biomass were incubated at 28°C and 31°C for 15 days. Juvenile *P. meandrina* with high biomass were six times more likely to die at 31°C versus 28°C, but corals with low biomass were 48 times more likely to die at 31°C versus 28°C. When juvenile *Pocillopora* were grown in seawater augmented with bicarbonate to reduce the cost of skeletogenesis in support of growth, growth was not affected, but energy expenditure was reduced by 20% to reduce reliance on metabolic reserves. Resource limitation of juvenile corals can affect their response to elevated temperatures, supporting the hypothesis that a physiological crisis initiated by resource limitation mediates the stringency of recruitment bottlenecks.

1. Introduction

The 'coral reef crisis' [1] is emblematic of ecosystem degradation caused by global climate change [2]. For coral reefs, quantification of the rate of recruitment of scleractinian corals is foundational to an understanding of the mechanisms driving the crisis, with evidence to date showing that the density of corals settling on standardized surfaces (i.e. tiles) has declined [3,4]. On the Great Barrier Reef, bleaching in 2016 and 2017 disrupted the stock-recruitment relationship for reef corals [5], which has critical implications for coral population dynamics [6,7] with effects determined in part by the post-settlement success of newly settled corals [8]. Experimental work in Moorea, French Polynesia, has demonstrated the importance of corallivory in determining the post-settlement success of corals [9,10] and mediating their distribution across depths [11]. In the Caribbean, brooding corals settle in relatively large numbers on shallow reefs, yet in recent decades they have not supported population growth [12]. Estimates of the number of juvenile corals produced by each settler as a proxy for post-settlement success has revealed the potential of this relationship to account for changes in coral cover [13].

The factors that kill juvenile corals remain unclear, especially in the context of warming oceans. Smaller corals often have higher mortality rates than larger corals [14] and larger corals have higher fecundity [15], which imposes selection for rapid growth to increase colony size [16]. However, faster growth incurs metabolic expenditures to synthesize tissue and deposit skeleton [17], which can lead to an energy deficit and the potential for death [18]. We addressed the role of metabolic reserves (estimated by measuring biomass) in

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mediating post-settlement success in corals, hypothesizing that the selective pressure to grow beyond small size classes [16,19] requires prioritizing the allocation of metabolic reserves to support rapid growth [17], assuming energy is a limiting resource. Consequently, there is the potential for an energy deficit when demands exceed supply, with extreme cases leading to the depletion of metabolic reserves posing a physiological crisis that few juvenile corals survive.

We conducted experiments in Moorea where an outbreak of crown of thorns seastars (*Acanthaster solaris*) followed by a cyclone left very low coral cover on the fore reef by 2010, yet by 2019 coral cover was 75% at 10 m depth [13]. High coral recruitment was implicated in rapid coral community recovery [20], making it compelling to ask what makes a coral recruit successful? We focused on spawning *Pocillopora* spp. and genetically identified individual species so that our results would not be confounded by species differences. We tested the hypotheses that: (i) a physiological crisis develops in juvenile corals through the depletion of tissue biomass in support of rapid growth, leaving them susceptible to mortality, and (ii) metabolic reserves are depleted through an energy deficit caused by the energy expenditures associated with the metabolic rate and the rate of development (i.e. the cost of development [21]). These hypotheses were tested by changing the biomass of corals and subjecting them to a thermal challenge with survivorship as the dependent variable. The mechanisms underlying the outcome of this experiment were explored by incubating corals in seawater with elevated bicarbonate concentrations to reduce the cost of growth (after [22,23]). We anticipated that these conditions would enhance growth and depress metabolic expenditures (i.e. lower respiration), thus easing the recruitment bottleneck.

2. Methods

(a) Survivorship

(i) Overview

In Phase I, temperature and day length were modified in an 18 day experiment to alter biomass through enhanced consumption of food reserves by high respiration and a decreased ability to replace them by photosynthesis. In Phase II, corals with low or high biomass were subjected to a 15 day thermal challenge during which survival was measured. All tanks were independently heated, chilled, illuminated and aerated.

(ii) Coral collection

On 3 and 4 April 2024, juvenile colonies of *Pocillopora* spp. 10–30 mm diameter (n = 240) were collected from the back reef (2–3 m depth). These small colonies probably were 2–3 years old and sexually immature. The corals were swabbed (Puritan HydraFlock®) to collect DNA for use in species identification, prepared as nubbins [24], and on 4 April placed in a 1000 l tank at 28.7 ±0.1°C (mean ± s.e., n = 13) and illuminated with LED lamps.

(iii) Phase I: biomass manipulation

The biomass of the corals was changed over 18 days by enhancing the consumption of metabolic reserves (a component of biomass) through elevated respiration at high temperature, and by reducing the restocking of metabolic reserves by partial starvation elicited by reducing photoautotrophy using a photoperiod of <12 h. Biomass was determined by fixing the corals in 5% formalin in seawater, decalcifying in 5% HCl in freshwater and drying to a constant weight at 60°C. On 7 April, the corals were randomly allocated to four 150 l tanks (approx. 50 corals tank^{-1}) supplied with filtered (approx. 150 µm) seawater from Cook's Bay (adjacent to the marine laboratory where the experiments were conducted) at approximately 120 ml min^{-1}. The tanks contrasted a target of 28.0°C (two tanks) on a 12 : 12 h photoperiod with 30.0°C on a 8 : 16 h photoperiod (two tanks), with light at a maximum photon flux density (PFD) of approximately 1200 µmol photons m^{-2} s^{-1}.

(iv) Phase II: thermal challenge

Following Phase I, the corals were subjected to a 15 day thermal challenge to contrast survivorship (the dependent variable) under target temperatures of 28.0°C (ambient summer) versus 31.0°C (thermal stress, elevated) and a 12 : 12h photoperiod at a maximum PFD of approximately 1200 µmol photos m^{-2} s^{-1}.

(v) Statistics

Analyses were completed using Systat 13 (Inpixon, San Jose, CA) and R (R version 4.4.1). To estimate how the relationship between biomass and area differed among treatments in Phase I, a gamma generalized linear mixed model (glmm) with a log link function was used. To compare survival times among colonies from Phase I that had 'high' or 'low' biomass and went into Phase II, Cox proportional hazards survival analyses were used. To evaluate whether coral sizes could confound treatment effects, sizes were compared among treatment combinations using a linear mixed effects model (with Phase II tanks as a random effect) for each taxon.

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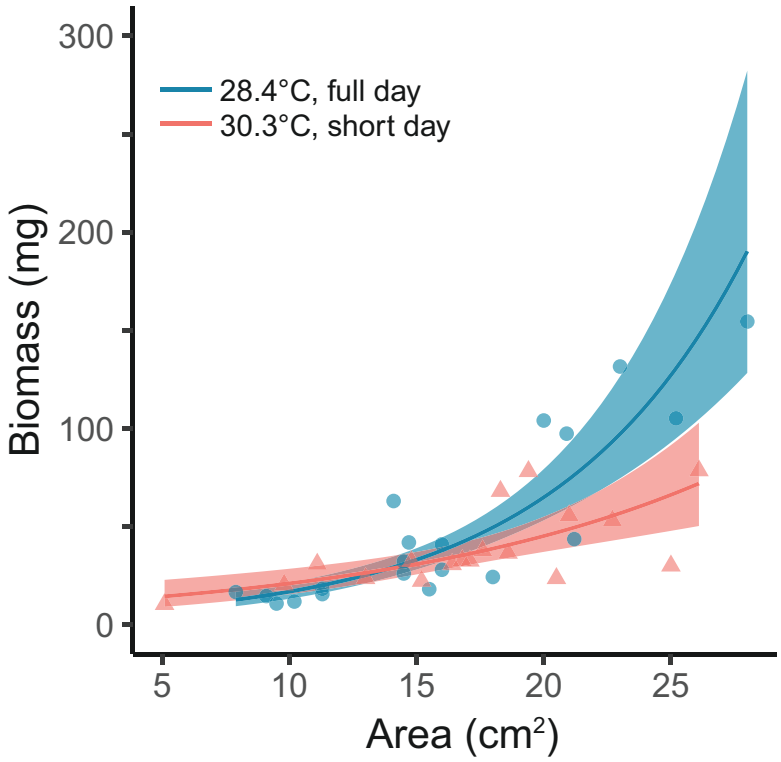

Figure 1. Biomass versus area for *Pocillopora* spp. incubated at 28.4°C on a 12 : 12 h (full day) photoperiod (light : dark), or 30.3°C on an 8 : 16 h (short day) photoperiod. Two outliers were removed (all data in electronic supplementary material, figure S1).

(b) Growth

(i) Overview

In August 2024, growth was stimulated by elevated bicarbonate concentrations in seawater [22,23] with the response measured through change in mass and dark aerobic respiration. Dark respiration provides an estimate of the summed metabolic energy expenditure for chemical and mechanical work, a portion of which is used for calcification and tissue synthesis [17,21]. This is likely to be an underestimate of the energetic cost of growth because it does not consider light-enhanced respiration [25].

(ii) Coral collection

Juvenile colonies (<30 mm diameter, $n = 70$ colonies) of *Pocillopora* spp. were collected on 9 August from 2–3 m depth in the back reef of Moorea and prepared as nubbins.

(iii) Treatments

On 10 August, corals were placed for 14 days in 80 l tanks supplied with approximately 600 ml min^{-1} of seawater under sunlight at a maximum of approximately 617 µmol photons m^{-2} s^{-1} (measured with a Li-Cor LI 193). There were two tanks per treatment that contrasted ambient seawater (for temperature, light, salinity and total alkalinity (A_T), A_T approximately 2300 µmol kg^{-1}) versus ambient seawater augmented with HCO_3^- (A_T~3200 µmol kg^{-1}). Seawater chemistry was evaluated by measuring pH and A_T (after [26]), from which carbonate chemistry was calculated using CO2SYS [27]. Coral growth (dependent variable 1) was measured through the change in dry mass over the 14 day experiment as calculated by change in buoyant weight (±0.001 mg), and aerobic respiration (dependent variable 2) was measured at the end of the experiment using oxygen respirometry.

(iv) Statistics

Seawater conditions in the four tanks were compared between treatments using a one-way nested PERMANOVA. Growth and respiration were analysed using a Gaussian linear mixed effects (LME) model.

For details of both experiments refer to electronic supplementary material.

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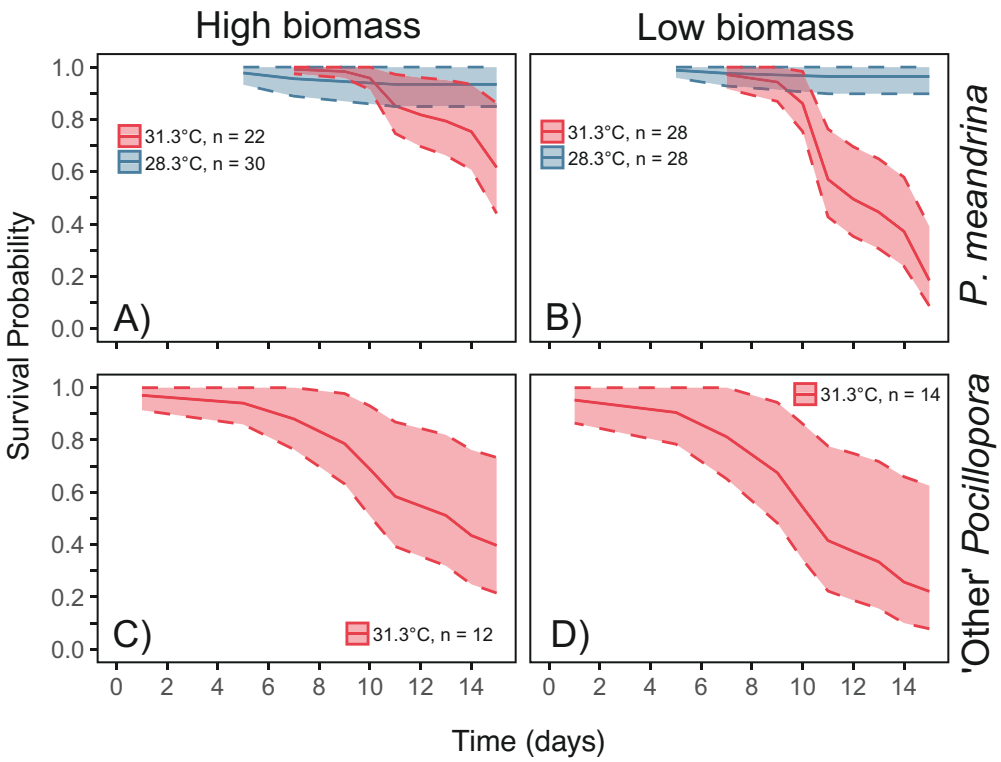

Figure 2. Survivorship of *Pocillopora* spp. exposed to a Phase I treatment altering biomass (figure 1) and a treatment testing for susceptibility to high temperature (Phase II). Solid lines are survival curves from Cox regressions with ± 95% confidence belts. (A,B) *P. meandrina* with a model including Phase I treatments and run separately for each Phase II treatment (28.3 and 31.3°C) with 22 to 30 corals per group. (C,D) 'Other' *Pocillopora* (12 colonies of *P. grandis*, nine *P. tuahiniensis*, three *P. cf. effusa*, one *P. verrucosa*, and one *P. acuta*) with a model that includes only the Phase II treatment of 31.3°C because in the 28.3°C treatment all corals with high biomass, and most of the corals with low biomass survived.

Table 1. Cox regression results for *Pocillopora meandrina* and 'other *Pocillopora*' exposed to an initial Phase I and a subsequent Phase II treatment.

		adjusted coefficient		adjusted hazard ratio (AHR)		
taxon	contrast	log(AHR)	95% CI	AHR	95% CI	*p*
Pocillopora meandrina	Phase I = low biomass, Phase II = 28.3 versus 31.3°C ($n = 56$)	3.87	1.86–5.89	48.13	6.43–360.12	<0.001
	Phase I = high biomass, Phase II = 28.3 versus 31.3°C ($n = 56$)	1.82	0.27–3.37	6.17	1.31–29.11	0.021
other *Pocillopora* spp.	Phase I = low versus high biomass, Phase II = 31.3°C ($n = 26$)	0.49	−0.44–1.42	1.63	0.64–4.14	0.302

3. Results

(a) Survivorship

(i) Overview

In April 2024, the mean (±s.e.) temperature in the lagoon was 29.4 ± <0.1°C (n = 30 days). Phase I contrasted 28.3 and 30.3°C (temperatures differed significantly, $F_{1,212} = 172.638$, $p < 0.001$) at a PFD of 1212 µmol photons m^{-2} s^{-1} (electronic supplementary material, table S1). The experiment began with 233 corals (electronic supplementary material, figure S1) with a mean (±s.e.) diameter of 25.6 ± 0.5 mm. Twenty corals were randomly selected and sacrificed for the determination of area, and 20 for biomass, with the treatments to alter biomass administered to 193 corals. Following Phase I, 40 corals were sacrificially sampled for biomass leaving 153 corals that had been genetically identified to species (four corals were lost to experimental error). These were transferred to Phase II that contrasted 28.3 and 31.3°C (that differed significantly, $F_{1,164} = 138.999$, $p < 0.001$) at a PFD of 1178 µmol photons m^{-2} s^{-1} (electronic supplementary material, table S1).

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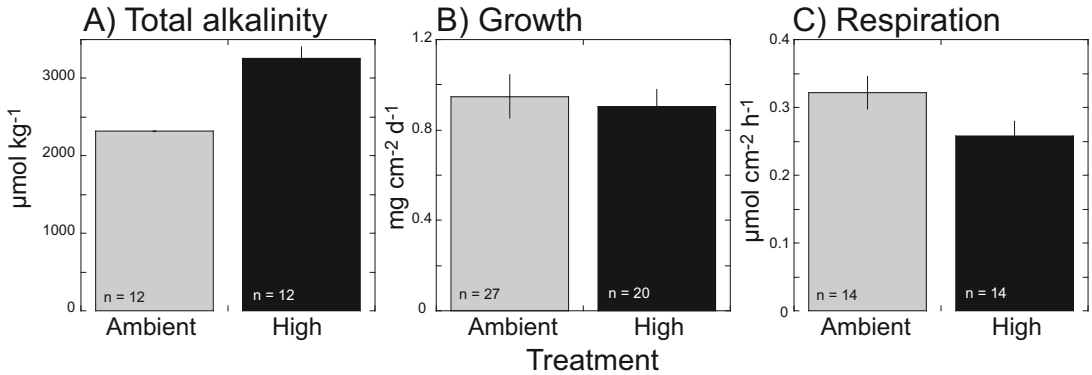

Figure 3. Effects of seawater A_T, ambient versus high (A), on the skeletal growth (B) and respiration of juvenile colonies of *Pocillopora* spp. Bars show mean ± s.e. with sample size (*n*).

(b) Phase I: biomass manipulation

For freshly collected corals, colony area increased with colony diameter ($F_{1,18}$ = 32.741, p < 0.001), allowing area to be calculated from diameter in the corals sacrificed for biomass; of these, 15 were *P. meandrina*, three were *P. grandis*, one was *P.* cf. *effusa* and one was *P. acuta*. Their mean (±s.e.) area-normalized biomass was 1.1 ± 0.1 mg cm^{-2}. Biomass (mg) increased with area ($F_{1,18}$ = 99.564, p < 0.001) (electronic supplementary material, figure S3A).

Following Phase I, biomass was quantified in 26 colonies of *P. meandrina*, four *P. tuahiniensis*, three *P. grandis*, two *P. verrucosa*, one *P. acuta*, one *P. cf. effusa*, and one of an unknown identity. For *Pocillopora* spp., the relationship between biomass and area differed between treatments (interaction: χ^2 = 6.32, d.f. = 1, p = 0.012) (figure 1, electronic supplementary material, S2). At 28.3°C and 12 h days, biomass increased by 14.38% (11.06–17.81, 95% CI) for every 1 cm^2 increase in area. At 30.3°C and 8 h days, the rate at which biomass increased with area declined by 5.66% (1.28–9.84, 95% CI). As biomass increased with area and the Phase I treatment had a proportional effect on biomass, the discrepancy in biomass between treatments increased with coral size. This made it statistically challenging to resolve the treatment effect on biomass in the smallest corals but it did not confound the biomass treatment in Phase II, as coral sizes did not differ between treatments (described below).

(c) Phase II: thermal challenge

Colony size did not differ among main effects (Phase I and Phase II treatments) or their interaction for *P. meandrina* ($F_{1,40}$ ≤ 0.570, p ≥ 0.396) and *Pocillopora* spp. ($F_{1,41}$ ≤ 0.889, p ≥ 0.737).

For *P. meandrina* with high biomass in Phase II, the first coral died on day 5 at 28.3°C (n = 30). Another died at 28.3°C on day 11, and eight died (36%) at 31.3°C (n = 22) from days 7 to 15 (figure 2A,B). For low biomass corals, death was more common, with one coral dying on day 7 at 28.3°C (n = 28), and 24 (86%) at 31.3°C (n = 28) from days 10 to 15 (figure 2A,B). A proportional hazards test confirmed the proportionality assumption for the model fit to corals with high (χ^2_1 = 1.88, p = 0.17) and low (χ^2_1 = 3.61, p = 0.06) biomass. On any day, 6.17 (1.31–29.11, 95% CI) times as many high biomass corals died at 31.3°C compared to 28.3°C, whereas 48.13 (6.43–360.12, 95% CI) times as many low biomass corals died at 31.3°C compared 28.3°C (table 1). For the other *Pocillopora* spp. in Phase II, none with high biomass died at 28.3°C (data not plotted), but at 31.3°C, one died on day 5, and eight between days 9 and 15 (figure 2C,D). The odds of mortality at 31.3°C did not differ between colonies with low and high biomass (table 1).

(d) Growth

(i) Overview

In August 2024, the mean (± s.e.) temperature in the backreef was 27.0 ± 0.1°C (n = 22 days). Biomass of the corals (electronic supplementary material, figure S1) increased with size (electronic supplementary material, figure S3B), with homogeneous slopes between April and August ($F_{1,26}$ = 0.293, p = 0.593), but a greater elevation in August versus April ($F_{1,27}$ = 11.090, p = 0.003) (electronic supplementary material, figure S3). The tanks were maintained at a mean (± s.e.) temperature of 28.6 ± 0.1°C (n = 4), a mean (± s.e.) maximum PFD of 408 ± 33 µmol photons m^{-2} s^{-1} (n = 4), and they contrasted a mean (± s.d., n = 2) A_T of 2319 ± 6 µmol kg^{-1} versus 3261 ± 1 µmol kg^{-1} that significantly differed (pseudo-$F_{1,45}$ = 120.55, $p_{\text{-perm}}$ = 0.001) with a 41% bicarbonate enrichment (electronic supplementary material, table S1, figure 3A).

Skeletal growth at ambient A_T ranged from 0.32 to 2.52 mg cm^{-2} d^{-1}, and under elevated A_T, from 0.25 to 1.68 mg cm^{-2} d^{-1}; these rates are not statistically different ($F_{1,43}$ = 0.210, p = 0.649, figure 3B). Dark respiration under ambient A_T ranged from 0.19 to 0.54 µmol O_2 cm^{-2} h^{-1}, and under elevated A_T from 0.18 to 0.50 µmol O_2 cm^{-2} h^{-1}; these rates differed (log transformed, $F_{1,24}$ = 5.281, p = 0.031, figure 3C). When A_T was elevated by 41% through bicarbonate augmentation, skeletal growth was sustained at a constant rate while metabolic expenditure was reduced by 20%, thus alleviating the requirement for respiratory substrates.

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4. Discussion

For tropical reef corals, it is widely accepted that changes in the survivorship of juvenile corals can create large differences in the number of corals growing into adult size classes [16,28], but little is known of the factors causing survivorship to vary. We tested the potential for physiological impairment to mediate post-settlement mortality in juvenile *Pocillopora* spp. We show that: (1) low biomass elevates coral mortality at high temperature, and (2) the metabolic cost of rapid growth likely causes biomass to decline through the catabolism of metabolic reserves. These findings suggest that a 'physiological crisis' can develop in young corals through the confluence of the selective pressure for small colonies to rapidly grow [16,19], the metabolic expenditures associated with rapid growth, and the need to support the expenditure of energy by catabolizing food reserves [21]. As reserves are consumed, juvenile corals either achieve escape in size (*sensu* [29]), so that the selective pressure for rapid growth attenuates and the energy deficit can be reconciled through feeding, or they succumb to damage they no longer have the metabolic energy to overcome.

Juvenile corals are exposed to strong selective pressure for rapid growth [16], because greater size improves the ability for spatial competition [30], the ability to maintain a positive energy balance [31], access to resources in the seawater above the momentum boundary later [32], and resistance to corallivory and mechanical damage [33]. In symbiotic corals, energy and resources to support growth come from autotrophy (supported by algal symbionts) and heterotrophy [34,35]. The metabolic expenditures associated with rapid growth are expressed through aerobic respiration that supplies ATP to meet the energetic requirements of chemical and mechanical work [36], particularly as they support calcification and tissue production [17]. When the demand for metabolic energy for rapid growth exceeds the supply from feeding [35], the energy deficit must be reconciled by metabolizing tissue reserves (i.e., lipids, proteins, and carbohydrates) [21,25] as respiratory substrates, or by reductions in metabolic expenditure (e.g., metabolic depression) [37]. The depletion of metabolic reserves reduces holobiont biomass as occurs towards the end of summer [38–40], and provides one means to address the short-term challenges of a deficit in metabolic energy. If the energy deficit cannot be corrected so that a reduction in biomass ensues, there are long-term implications of this trend including vulnerability to the metabolic costs of wound repair [41], impaired reproduction [42], and enhanced mortality during exposure to environmental challenges [43–45]. For juvenile *Pocillopora* spp. in Moorea, we reduced their biomass by increasing respiration at elevated temperature [40,46], and lowered photosynthesis through a reduction in integrated light [47] over shortened days [48], which also is likely to reduce the potential for autotrophy [40,46].

Other factors that affect the probability of small corals dying [49] include corallivory [9,10], spatial competition with algae, sponges and other taxa [50], bleaching [51,52], pathogens [53], or feeding on plankton [54]. While small corals can benefit from their size by exploiting shading from adjacent organisms to reduce their susceptibility to bleaching [55,56], as well as associational refuges with algae to reduce their exposure to fish corallivory [57], the overall drawbacks of being small place great value in 'escape in size' [29,58,59]. Small corals inevitably are embedded in the momentum boundary layer above the benthos [32], which limits access to metabolites and heterotrophic food sources, and while shading associated with small size can have beneficial consequences (described above), it can also reduce the autotrophic benefits of algal symbionts [34].

When juvenile *P. meandrina* were exposed to high temperature, corals with low biomass died faster than those with high biomass (figure 2). The likely cause of this effect is reduced size of food reserves and fewer tissue resources with which the corals can respond to environmental challenges, including in the present case, the costs of damage attributed to elevated temperature [60]. The reduced mortality of *P. meandrina* with high biomass (figure 2A,B) is consistent with the hypothesis that a 'physiological crisis' mediates the stringency of the recruitment bottleneck, although the results for 'other *Pocillopora*' suggests that expression of the 'physiological crisis' may differ among species. To evaluate the role of metabolic expenditures in this crisis, we sought to increase the ease with which corals could grow, rationalizing that conditions suitable for growth would reduce the metabolic expenditure on this process and alleviate the catabolic depletion of food reserves [25].

Bicarbonate additions to seawater can stimulate coral calcification [22,23], most likely through the passive delivery of HCO_3^- through paracellular pathways [61] to the sites of use for photosynthesis and calcification [62]. The exact mechanisms by which HCO_3^- could alleviate the metabolic expenditure associated with growth are unknown, but elevated HCO_3^- concentrations would reduce the need for carbonic anhydrase to convert metabolic CO_2 into HCO_3^- in support of calcification [62]. Bicarbonate could also enhance calcification through the removal of protons from the extracellular fluid by stimulating photosynthesis, and promoting the influx of calcium ions to support calcification [62]. We found that HCO_3^- augmentation did not enhance calcification in juvenile *Pocillopora*, but it was associated with a 20% reduction in dark aerobic respiration (i.e. metabolic expenditures were reduced). It is unclear why HCO_3^- did not promote calcification (cf. [22,23]), but it is possible these early life stages of corals lack the metabolic architecture to use enhanced HCO_3^- access to accelerate calcification. Elucidating the mechanism underlying our results for HCO_3^- enrichment will require further work. Nonetheless, our results suggest that conditions promoting calcification and photosynthesis [22,23] allow juvenile corals to maintain growth rates with lower metabolic expenditure. This would reduce the demand for food to support metabolism, slow the rate at which biomass would need to be catabolized to balance the energy budget, and favour biomass retention in juvenile corals to enhance their resistant to environmental stress.

The results of our experiments suggest that rapid growth in juvenile corals incurs high metabolic expenditure, which favours the depletion of metabolic reserves and enhanced susceptibility to stress through starvation [63]. In concert with corallivory [11] and spatial competition [50], the onset of a physiological crisis in juvenile corals increases the stringency of the recruitment bottleneck preventing them from becoming adults. Alleviation of this bottleneck, for example through enhanced feeding of early coral life stages [64,65], could be a promising target for human intervention to promote coral community recovery following disturbances.

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Ethics. All applicable international, national and/or institutional guidelines for the care and use of animals were followed. The research was conducted under permits issued by Gourvernement de la République en Polynésie Francaise (DTRT) (Convention D'Accueil, 2024) and DIREN permits 2907 and 3936.

Data accessibility. Data are available from the Dryad Digital Repository [66].

Supplementary material is available online [67].

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. P.E.: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, resources, supervision, validation, visualization, writing—original draft, writing—review and editing; A.C.: investigation, methodology, validation, writing—review and editing; S.B.: conceptualization, formal analysis, methodology, software, validation, visualization, writing—original draft, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration. We declare we have no competing interests.

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